

EX2 - T115952129

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2026-10-05

Data

```
commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38, 29, 24,
            31, 36, 40, 26, 33, 28, 21, 44, 30, 37, 23, 39)
```

Question 1

```
max(commute)
```

```
## [1] 45
```

```
min(commute)
```

```
## [1] 18
```

```
range <- max(commute) - min(commute)
class_width <- range / 5
```

```
table(cut(commute, 5))
```

```
##
##  (18,23.4] (23.4,28.8] (28.8,34.2] (34.2,39.6] (39.6,45]
##           5           5           5           5           4
```

```
frequency <- table(cut(commute, 5))
cumulative_frequency <- cumsum(frequency)
cumulative_frequency
```

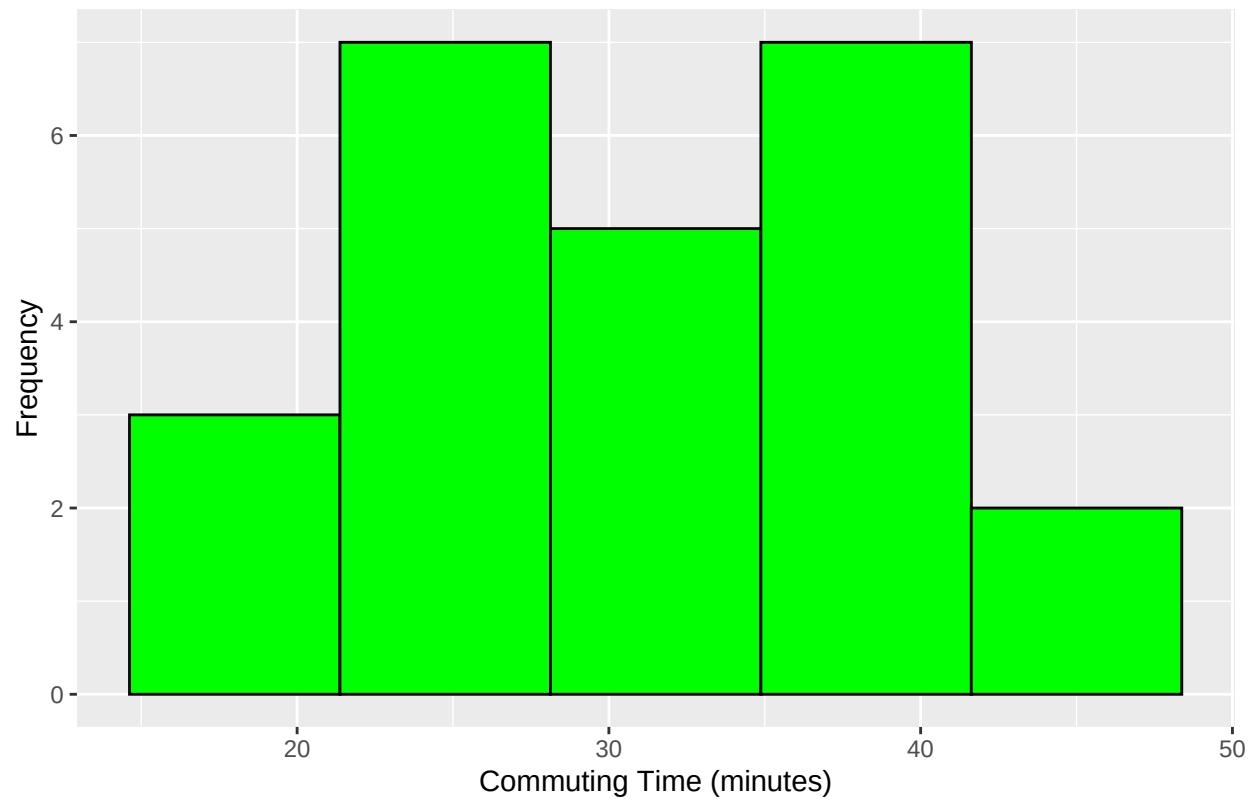
```
##  (18,23.4] (23.4,28.8] (28.8,34.2] (34.2,39.6] (39.6,45]
##           5           10           15           20           24
```

Question 2

```
library(ggplot2)
```

```
ggplot(data = data.frame(commute = commute), aes(x = commute)) +
  geom_histogram(bins = 5, color = "black", fill = "green") +
  labs(
    title = "Histogram of Daily Commuting Times",
    x = "Commuting Time (minutes)",
    y = "Frequency"
  )
```

Histogram of Daily Commuting Times



Question 3

```
upper_boundaries <- c(23.5, 29.5, 35.5, 41.5, 47.5)
cumulative_frequency <- c(5, 11, 16, 22, 24)
ogive_data <- data.frame(
  upper_boundaries,
  cumulative_frequency
)
ggplot(ogive_data, aes(x = upper_boundaries, y = cumulative_frequency)) +
  geom_line() +
  geom_point() +
  labs(
    title = "Ogive of Daily Commuting Times",
    x = "Upper Class Boundary (minutes)",
    y = "Cumulative Frequency"
  )
```

Ogive of Daily Commuting Times

